

Module 04: Cognition in Robotics — Autonomous Reasoning

Learning Objectives

1. Define **autonomous cognition** as the capacity of a robot to reason about its situation, assess its own competence, and make strategic decisions without human guidance.
2. Analyze how **situation assessment, meta-cognition, and goal management** implement higher-order Active Inference in autonomous systems.
3. Apply cognitive architectures to explain how autonomous robots handle novel situations.

Introduction

Autonomous cognition goes beyond state estimation (where am I?) to include situation assessment (what is happening?), self-assessment (am I capable of handling this?), and strategic reasoning (what should I prioritize?). This is the "thinking" that distinguishes a truly autonomous agent from a mere reactive system.

Key Concepts

1. Situation Assessment

Situation assessment constructs a high-level interpretation of the environment:

- **Scene classification:** Is this a normal operating environment, a hazardous zone, or an emergency?
- **Event detection:** Has something significant happened (object collision, human fall, door opening)?
- **Temporal reasoning:** How has the situation evolved over time? Is it getting better or worse?

In Active Inference terms, situation assessment is inference at a **high level of the generative model hierarchy** — abstracting from low-level percepts to categorical state descriptions that support strategic decision-making.

2. Meta-Cognition: Monitoring One's Own Model

Autonomous agents need **meta-cognition** — the ability to assess the reliability of their own inference:

- **Confidence monitoring:** How certain am I about the current state estimate? If confidence drops below a threshold, the agent should switch to more conservative behavior.
- **Novelty detection:** Is the current situation similar to anything in the training data? If not, the generative model may be unreliable and the agent should seek information or request help.
- **Competence assessment:** Can I achieve this goal with my current capabilities? If the EFE for all available policies exceeds a threshold, the task may be beyond the agent's competence.

Meta-cognition is Active Inference about Active Inference — a higher-order generative model that predicts the reliability of the lower-order model.

3. Goal Management and Priority

Autonomous agents typically have multiple concurrent goals that require prioritization:

- **Safety goals** (don't collide, don't damage, don't harm) always take highest priority
- **Task goals** (deliver packages, assemble widgets, patrol area) are the agent's primary mission
- **Maintenance goals** (recharge battery, calibrate sensors, update map) ensure continued operability
- **Social goals** (don't block hallways, don't startle people, communicate status) enable coexistence

Goal management requires a **priority architecture** where higher-priority goals can preempt lower-priority actions — precisely the hierarchical policy selection of Active Inference.

4. Anomaly Detection and Explanation

When something unexpected happens, autonomous agents should both detect the anomaly and generate an explanation:

- **Detection:** A spike in prediction error across multiple channels simultaneously signals an anomaly — something the generative model did not predict
- **Classification:** Is this anomaly dangerous (obstacle in path), benign (new decoration in office), or informative (clue about environmental change)?
- **Explanation generation:** For human-supervised systems, the robot should communicate *why* it detected an anomaly — providing interpretable reasoning that enables human oversight

Applications

- **Hospital delivery robot:** An autonomous hospital robot uses situation assessment to navigate complex social environments — classifying hallway traffic as "normal flow," "emergency rush," or "patient transport" and adjusting its behavior accordingly. Meta-cognition monitors localization confidence, triggering map recalibration when the hospital reconfigures.
- **Inspection drone with anomaly reporting:** A drone inspecting infrastructure detects anomalies (cracks, corrosion, deformation) through prediction error on its structural model, classifies severity, and generates explanatory reports with annotated images for human engineers.

Conclusion

Autonomous cognition — situation assessment, meta-cognition, goal management, and anomaly detection — implements higher-order Active Inference that enables strategic, self-aware robotic behavior. The next module examines autonomous action.